

Physiology and Behavior of the Hen During Induced Molt

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ABSTRACT Feed deprivation has been adopted by the commercial egg industry to induce molt because it is the easiest method to apply and produces the best results. Feed deprivation, however, raises concerns about animal welfare. Birds respond to long-term feed deprivation in three phases. The first phase lasts at most a few days, during which physiological and behavioral adjustments ultimately reduce protein catabolism and energy expenditure. A temporary increase in plasma corticosterone may be observed at this time. Corticosterone promotes gluconeogenesis, helping to maintain plasma glucose levels in the initial stage of the fast. The corticosterone increase may also be linked to increased activity in feed-deprived birds. Hens have been observed to manifest temporarily increased levels of alertness and activity during the first 48 h of feed deprivation. Aggressive behavior of hens also has been observed to increase briefly during the first day of feed deprivation. The second phase is the longest, during which proteins are spared and lipids are catabolized to provide energy. This phase may last several months in some species; in the chicken it can continue

more than 20 d. Hens show increasing amounts of resting behavior during this phase. The third phase begins when protein catabolism accelerates. A pathological stage eventually is reached when the bird will cease activity and no longer eat. The phased response to feed deprivation optimizes a tradeoff between the need to maintain constant levels of plasma glucose to sustain activity and the need to preserve critical body structures such as muscles and organs. Hens are capable of vigorous activity throughout feed deprivation periods typical of induced molts, which do not appear to take birds beyond the second phase of fasting. Hens having undergone extended fasts may also have improved livability. Alternative induced molting methods are being sought to reduce animal welfare concerns. The methods of current interest involve alteration of feeding regimen and cause at least some body weight loss. These alternative methods should be evaluated to ensure that they do not actually make aspects of hen welfare worse compared to feed withdrawal, which might happen if hens perceive feed restriction without being allowed to progress fully into the second phase of adaptation to feed deprivation.

(*Key words:* fasting physiology, fasting behavior, feed deprivation, feed restriction, induced molt)

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INTRODUCTION

Molt induction to rejuvenate the egg laying performance of commercial laying hen flocks is an important practice in many parts of the world, often being necessary to make a flock profitable under certain market scenarios. The first objective of an induced molt program is to cause hens to cease egg production and enter a non-reproductive state. The programs used to achieve this objective in commercial flocks may involve feed deprivation for 14 d or more in warm environments and typically cause BW losses of 25 to 35%. Such programs raise concern about animal welfare because it is thought that they may be harmful to hens.

METHODS TO INDUCE MOLT

Baker et al. (1983) looked at the relationship between BW loss and postmolt production performance of caged layers and came to the conclusion that a weight loss of 30% produced the optimum subsequent performance in terms of egg production and shell quality. Because feed withdrawal is easy to implement, feed withdrawal to achieve BW loss of 25 to 35%, followed by a rest period and then restimulation of egg production has become the typical industry method for induced molting. Many scientists have investigated alternatives that might be more humane. These alternatives fall into two general categories:

1. Various methods of nutrient restriction that avoid long-term feed withdrawal
2. The use of dietary additives.

Feed or Nutrient Restriction

In the former category, the simplest approach has been to remove feed for shorter periods. Hens fasted

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for 4 d can have comparable postrest egg production performance to those fasted for 10 d, although egg specific gravity, which is a measure of the amount of shell on an egg, tends to be higher for hens fasted longer (Christmas et al., 1985; McCormick and Cunningham, 1987; Koelkebeck et al., 1992). Cunningham and McCormick (1985) found that a 10-d feed removal program gave better economic returns than a 4-d program over the course of two postmolt production cycles. Another approach has been to alternate periods of feed withdrawal with days of feeding, or to limit the amount of feed provided per day, until the target BW loss is achieved (Zimmermann et al., 1987; Zimmermann and Andrews, 1990; Rolon et al., 1993; Buhr and Cunningham, 1994). Targeted weight losses have ranged from 15 to 35%. Generally speaking, the length of the recurring period without feed (up to 5 d) and the amount of BW loss in these studies did not significantly affect egg production over the second production cycle. Within the range of BW losses tested, hens with lower weight losses tended to produce more eggs early in the postmolt cycle. Conversely, those with the greater weight losses produced more eggs in the latter part of the postmolt cycle. The molt programs causing greater BW losses did not necessarily produce higher egg specific gravity. Zimmermann et al. (1987) noted that molt programs involving a series of short periods of feed withdrawal conserved feed over a program with a single period of feed withdrawal to the same target BW loss.

Calcium is an important element for maintenance of egg production in the chicken, not only because large amounts of it are necessary for shell formation, but also because it plays a role in the secretion of gonadotrophic hormones (Luck and Scanes, 1980; Decuyper and Verheyen, 1986; Brake, 1993). Attempts have been made to arrest egg production of hens by feeding diets low in calcium (Douglas et al., 1972; Gilbert and Blair, 1975; Hurwitz et al., 1975; Rolon et al., 1993). These were effective at reducing egg production, with the diets lowest in calcium causing the lowest rates of production, but did not cause flocks to cease production entirely. Even at very low levels of dietary calcium, the ovary and oviduct did not regress to a non-reproductive state (Douglas et al., 1972; Gilbert and Blair, 1975). Peak egg production tends not to be as high after rests induced by low-calcium diets as after rests induced by feed deprivation, but some improvement in shell quality can be achieved (Hurwitz et al., 1975; Rolon et al., 1993). A problem regarding diets very low in calcium is that temporary paralysis (Douglas et al., 1972) or osteoporosis (Hurwitz et al., 1975) can occur. Hughes and Wood-Gush (1973) observed that calcium deprivation causes increased activity and pecking in pullets.

Diets deficient in sodium have been used to induce molt (Berry and Brake, 1987). This method may not cause hens to completely cease egg production and gives inferior second cycle performance. Hughes and Whitehead (1979) reported increased cannibalistic pecking among hens fed low sodium diets.

Dietary Metals

Dietary zinc is effective at inducing rests from egg production, with postrest production performance being similar to that of induced molt programs involving feed withdrawal (Shippee et al., 1979; Cunningham and McCormick, 1985; Goodman et al., 1986; Berry and Brake, 1987). Zinc has an inhibitory effect on ovarian function (Johnson and Brake, 1992), and when incorporated at relatively low levels in low calcium diets it causes oviposition to cease without greatly depressing feed consumption (Breeding et al., 1992). Increasing levels of dietary zinc, however, cause progressive declines in feed consumption such that hens given 20,000 ppm zinc in an otherwise typical layer ration virtually begin fasting (McCormick and Cunningham, 1984, 1987).

Dietary aluminum has also been tried as a molting agent (Hussein et al., 1989). The procedure can take more than 2 wk to cause egg production to cease, but production performance during a 14-wk second cycle comparison was similar to that of hens molted by feed withdrawal. Dietary aluminum also causes reduced feed intake by hens.

Enheptin

The drug enheptin when fed in the diet can induce molt in hens (Pino, 1955; Robblee and Clandinin, 1955). Dietary enheptin reduces feed consumption of hens and appears to cause dose-related BW loss.

Dietary Iodine

High levels of dietary iodine have been shown to induce molt in hens (Wilson et al., 1967). Second-cycle production performance was inferior to the nonmolted control in this study.

Hormone Agonist and Hormones

Dickerman and Bahr (1989) and Burke and Attia (1994) were able to induce molt in hens using formulations of leuprolide acetate, a gonadotrophin-releasing hormone agonist, delivered by implanted osmotic pump or by injection. Weight loss of the treated hens was little more than that due to regression of the reproductive organs. Improvements in egg production and shell quality were achieved, but full second-cycle performance comparisons against traditional molt procedures induced by feed deprivation were not made. The reproductive hormone, progesterone, when injected or fed in the diet will cause late-cycle hens to go out of production (Adams, 1955, 1956; Shaffner, 1955; Wilson et al., 1967). When progesterone is injected, the process is fast, but when the hormone is fed it tends to be slow to take effect. Improvement in egg production appears to be possible, but the use of progesterone as a molting agent for commercial use would require further study in a modern scenario.

The thyroid hormone, thyroxine, has been shown to terminate egg production and induce molting of hens (Sekimoto et al., 1987). The procedure involved daily injections of hens. Second cycle egg production performance was not evaluated. Infusion of the adrenal hormone, corticosterone, causes rapid ovarian and oviduct regression, apparently by reducing secretion of luteinizing hormone (Etches et al., 1984; Williams et al., 1985). Second-cycle egg production was not investigated. Dietary corticosterone has been found to produce levels of plasma corticosterone similar to those of the infusion studies within 1 d (Davison et al., 1983; Gross and Siegel, 1983). Dietary corticosterone does not cause reduced feed intake (Gross et al., 1980).

Any use of a hormone agonist or hormone to induce molt that involves injection or osmotic pump implantation would require skilled handling of each hen and would add substantial labor costs to an induced molt program, in addition to the price of the substance itself. It would also be necessary to verify that none of these substances are deposited into eggs. Although administration of corticosterone in feed might be simple, dietary corticosterone produces physiological responses in chickens similar to those produced by external stressors (Davison et al., 1983; Gross and Siegel, 1983) and causes decreased feed efficiency and increased liver and body fat (Gross et al., 1980; Siegel et al., 1989).

PHYSIOLOGY DURING FEED WITHDRAWAL

Because feed withdrawal is common industry practice for inducing molt, the remainder of this paper will summarize a variety of information pertaining to the responses of birds to feed deprivation. This summary will be useful as a basis for comparison when information on the responses of hens to alternative methods of molt induction becomes available.

Phases of Fasting

The existing scientific literature does not summarize the responses of domestic fowl to feed deprivation. Cherel et al. (1988), however, reviewed long-term fasting of birds in general from studies that reported different measures for a variety of avian species. They concluded that responses to fasting are similar across species and fall into three distinct phases.

Phase 1, the initial adaptation phase, is short, no more than a few days, regardless of species, and involves metabolic transition to a stable fasting physiological state. During this phase, there is rapid decrease of body mass but also a rapid reduction in rate of body mass loss. The basal metabolic rate declines. Peripheral body temperature decreases, but core body temperature does not decline in larger birds, i.e., species that do not exhibit torpor. Plasma corticosterone concentration increases. Hepatic glycogen disappears rapidly, and gluconeogen-

esis is established. Nitrogen excretion decreases, indicating a decline in rate of protein catabolism.

Phase 2 is a phase of long-term economy and may last weeks or even months, depending on the species. During this time, specific daily body mass loss, i.e., daily mass loss per unit of remaining body mass, is almost constant—if anything declining slightly as the phase proceeds. Specific basal metabolic rate is almost constant. Locomotor activity is reduced to conserve energy. Plasma corticosterone concentrations are low. Almost all energy expenditure is derived from fat catabolism, e.g., 93 to 94% in domestic geese (Le Maho et al., 1981). Protein catabolism is minimal, indicating that proteins are spared in this phase, and because loss of muscle mass is limited, vigor is maintained.

Phase 3 may last days to weeks, depending on the species (Cherel et al., 1988). A rise in rate of specific body mass loss occurs due to increased protein catabolism. The transition from Phase 2 to Phase 3, therefore, is characterized by the fact that proteins are no longer spared, and the fasting bird experiences accelerated loss of muscle mass. Significant lipid reserves remain at the start of the phase and fat catabolism continues concurrent with protein catabolism to supply energy needs. The specific basal metabolic rate does not change. Vigor continues until late in Phase 3. For example, fasting emperor penguins usually leave their colony coincident with the start of Phase 3 and walk up to 200 km to the sea to begin refeeding (Cherel et al., 1988). Refed birds recover condition and muscle mass without evident harm until late in the phase. Eventually, however, lipid reserves become depleted, and muscle loss from protein catabolism reaches a point at which debilitation occurs. The fasting bird will become motionless and cannot be refed, beginning an irreversible progression to death.

Body Mass Loss and Survival

Body mass loss of laying hens during long-term feed deprivation follows the pattern outlined by Cherel et al. (1988). Recalculations of BW data presented in Webster (2000) indicate that Phase 1 lasted about 3.5 d during which specific body mass loss averaged 47 g/kg per d. Thereafter, Phase 2 continued for the remainder of the feed-deprivation period (18 d) with specific body mass loss averaging 15 g/kg per d during the first half of the period and 13 g/kg per d during the latter half. Total weight loss was 35% of the starting BW. There was no increase of specific body mass loss rate to suggest transition to Phase 3. Similarly, in a study done more than a century ago, nitrogen excretion of feed-deprived domestic fowl was low and constant, consistent with the protein sparing of Phase 2, until the 24th d of starvation (Schimanski, 1879). Nitrogen excretion increased after d 24, coincident with increased rate of specific body mass loss. The fowl were reported to retain vigor for the first week of Phase 3 but became motionless after the seventh day (d 31).

Bierer et al. (1965, 1966) observed that survival time and ability of chickens to sustain weight loss during starvation increased with age. Eight-week-old broiler-type birds showed substantial ability to handle weight loss, surviving until they lost an average of 50% of their starting BW. Forty-week-old light breed hens lived a minimum of 30 d (average 38 d), consistent with the observations reported by Schimanski (1879). The mature hen evidently is able to cope with feed deprivation for considerably longer than the periods imposed during commercial induced molts.

Mortality of healthy hens subjected to feed deprivation typical of induced molt programs is usually not greater than for hens left on full feed (Berry and Brake, 1987; Koelkebeck et al., 1992; Webster, 1995; Alodan and Mashaly, 1999). Webster (2000) found that hens induced to molt by feed withdrawal had better survival during the 40-wk second production cycle than did control hens. Lee (1982) and Zimmermann et al. (1987) also reported improved livability of molted hens. In the commercial egg industry, mortality typically increases somewhat during induced molts and then is reduced for many weeks thereafter (Bell, 2003). Invariably, there are moribund hens in large commercial layer flocks. These normally die at a steady weekly rate. Induced molt programs probably hasten the death of moribund hens, compressing a certain total amount of mortality into a shorter time.

Plasma Glucose

In chickens, plasma glucose concentrations generally decline somewhat at the beginning of feed restriction or fasts, particularly in young birds (Nir et al., 1975; Freeman et al., 1980, 1983; Scanes et al., 1980; Katanbaf et al., 1989; Edwards et al., 1999), although Scanes et al. (1980) did not observe such a decline in adult birds. Within 24 h of starvation, glycogenolytic activity in the chicken liver greatly increases, promoting net glucose efflux from the liver, associated with a 90% reduction of liver glycogen (Davison and Langslow, 1975). Thereafter, even relatively young chickens engage metabolic mechanisms to produce glucose at a constant rate per kilogram up to 8 d of fasting (Brady et al., 1978). The feed deprivation treatment in the study by Brady et al. (1978) was more rigorous than its length might suggest because the weight loss of the pullets amounted to 36% of their starting BW, none of which could be attributed to loss of reproductive organ mass because the birds were not sexually mature. Brake and Thaxton (1979) found that plasma glucose concentrations actually increased in hens during periods of feed and water withdrawal. Edwards et al. (1999) reported only a slight decline in muscle glycogen levels at 24 h of feed withdrawal in female 6-wk-old broilers, with the glycogen levels returning to normal thereafter. The authors concluded that muscle glycogen is resilient to the metabolic changes that occur during feed deprivation. The available evidence for domestic fowl, therefore, corroborates

the assessment by Cherel et al. (1988) that fasting birds maintain euglycemia during long-term fasts.

Plasma Corticosterone

Feed withdrawal or feed restriction causes increased concentrations of plasma corticosterone in immature chickens (Freeman et al., 1980; Harvey and Klandorf, 1983; Mench, 1991). Age or experience appears to reduce this response to food shortage. Freeman et al. (1981) found that plasma corticosterone concentrations were elevated in feed-restricted chickens at first, but that these concentrations declined to equal those of full-fed chickens by 7 wk of age. Freeman et al. (1983) concluded that plasma corticosterone concentrations of 16-wk-old chickens were not greatly affected by feed withdrawal for 24 h. On the other hand, Savory et al. (1996) and Hocking et al. (1996) observed that, although plasma corticosterone levels declined with age in growing feed-restricted broiler breeders, these levels had a negative relationship with BW (i.e., the degree of growth suppression due to feed restriction) regardless of age. The birds that were more restricted from feed, thus smaller, had higher plasma corticosterone levels at any given age. All rations in these two studies provided sufficient nutrients and energy to support growth. For adult birds, Scanes et al. (1980) reported that plasma corticosterone rose during a 24-h fast of individually caged chickens, and Hocking et al. (1998) reported that plasma corticosterone was elevated in feed-restricted mature broiler breeder hens.

Zulkifli et al. (1995b) found that bird experience can alter the plasma corticosterone response to feed restriction. An early period of feed limitation eliminated subsequent plasma corticosterone increase in response to feed withdrawal in growing chickens.

Immobilization by hand or crating of a chicken causes much higher concentrations of plasma corticosterone than feed withdrawal (Beuving and Vonder, 1978). The plasma corticosterone rise in response to feed withdrawal is not extreme compared to that which can be induced by frightening stimuli. Psychological effects directly related to feed deprivation can also influence plasma corticosterone levels of domestic fowl. Harvey et al. (1983) found that mere sight of food was sufficient to cause plasma corticosterone levels to quickly decline in young chickens deprived of food for 48 h. After a few hours, however, plasma corticosterone settled to levels that were inversely related to the energy level of the feed provided.

Plasma corticosterone appears to be linked to bird activity in feed-deprivation situations. Hocking et al. (1996) noted that plasma corticosterone levels and activity levels in broiler breeders were higher when the degree of feed restriction was greater. In addition to being more active, the lighter, more feed-restricted birds were also less fearful. Astheimer et al. (1992) found that fasting passerine birds given exogenous corticosterone had increased activity levels. The same effect was not evi-

dent in fed passerines. The authors suggested that corticosterone might be important for initiation of food-searching behavior during periods of food deprivation.

There has been very little study of the plasma corticosterone levels of hens during the feed withdrawal phase of induced molts. Hoshino et al. (1988) reported that plasma corticosterone levels rose during the feather loss stage of an induced molt, but corticosterone was not measured during feed deprivation. Etches et al. (1984) molted hens such that feed and water withdrawal were alternated with periods of whole oats and water being provided ad libitum. Plasma corticosterone levels peaked 48 h after the start of the molt induction treatment and declined thereafter, similar to the general pattern outlined by Cherel et al. (1988) for phases 1 and 2 of a fast.

Heterophil:Lymphocyte Ratios and Immune Responses

Glucocorticoids, such as corticosterone, can cause changes in circulating populations of blood leucocytes by suppressing proliferation and activity of lymphocytes (Munck et al., 1984). It might be expected that changes in plasma corticosterone in chickens due to feed deprivation would lead to changes in numbers of heterophils and lymphocytes. Limited feeding can increase heterophil:lymphocyte (H:L) ratios of young broiler breeders (Maxwell et al., 1992; Zulkifli et al., 1994a; Hocking et al., 1996). This effect in some cases may be eliminated by habituation; Zulkifli et al. (1993) observed that H:L ratios returned to normal 12 d after the start of a feed-restriction program. Adolescent or adult birds on feed-restriction programs for long periods do not necessarily have higher H:L ratios than their respective unrestricted controls (Katanbaf et al., 1987; van Niekerk et al., 1988; Savory et al., 1992, 1996; Hocking et al., 1998). Complete withdrawal of feed for 2 d increases H:L ratios in immature and adult chickens (Wolford and Ringer, 1962; Gross and Siegel, 1983), but again, habituation appears to have a mitigating effect (Gross and Siegel, 1986). Broiler breeders fed only every second or third day had elevated H:L ratios at the end of the period without feed at 7 wk of age but not so at 21 wk of age (Katanbaf et al. 1989), suggesting that age might also play a role in the leucocytic response to feed deprivation.

A psychogenic influence on H:L ratios can occur in feed-deprivation situations. Zulkifli et al. (1995a) found that White Leghorn hens deprived of feed for 48 h had elevated H:L ratios, but the degree of elevation was less if the hens were provided sand in the feeder than if the feeder was removed entirely. H:L ratios were highest when feed was left in the feeder but access was blocked by wire mesh. Evidently, the feeder with sand moderated hens' psychological reactions relative to the situation involving total removal of the feeder. Thwarting of eating, on the other hand, exacerbated their reactions. Medvedev et al. (2002) found that hens molted with

alfalfa diets had H:L ratios intermediate to those of full-fed controls and hens molted by feed withdrawal; a result that perhaps parallels the result Zulkifli et al. (1995a) obtained with their sand treatment. It is possible in the study by Medvedev et al. (2002), however, that the respective H:L ratios were determined by differences in dietary energy consumed by hens in the different treatments and not by psychogenic influences related to characteristics of the treatments themselves.

Blood leucocyte populations generally have been found to change in hens during the feed-withdrawal periods of induced molt programs. Brake et al. (1982) found no significant alteration in numbers of lymphocytes during fasts used to induce molting of hens, and only in their first trial did they observe a significant increase in heterophils at the middle and end of the feed-withdrawal period. However, my own calculations using the data published, albeit unsupported by statistical analysis, suggest H:L ratios for fasted hens were higher on d 7 and 14 of trial 1 and on d 7 of trial 2 than for their respective controls.

Unlike Brake et al. (1982), Holt (1992) studied leucocytes early in feed withdrawal periods used to induce molts and found that lymphocytes were reduced in number on d 2 and 8 in experiment 1 and on d 1 in experiment 2. There were no other changes in leucocyte populations. Davis et al. (2000) and Medvedev et al. (2002) reported increased H:L ratios around d 7 of feed withdrawal during induced molt programs. Alodan and Mashaly (1999) and Kogut et al. (1999) tracked changes in heterophil populations in White Leghorn hens during 10- and 14-d periods of feed deprivation, respectively, finding that heterophil numbers increased by 2 d after feed withdrawal and remained elevated for the remainder of the feed-deprivation periods. Kogut et al. (1999) also found that heterophils from feed-deprived hens had reduced ability to phagocytize *Salmonella enteritidis* bacteria and to generate an oxidative burst in response to inflammatory agonists.

Ben Nathan et al. (1977) found that the weight of lymphoid tissue and antibody titers to *Escherichia coli* and sheep red blood cells were reduced in young chickens that had been feed and water deprived for 48 h. Holt (1992) observed that antibody responses to sheep red blood cells were unaffected by feed withdrawal to induce molts of hens, but titers to *Brucella abortus* were reduced for feed-deprived hens in his second experiment, and delayed-type hypersensitivity response, an indicator of cellular immune responsiveness, was depressed during feed withdrawal. Although corticosterone was not measured in these studies, the results are consistent with the immunosuppressive and anti-inflammatory actions of corticosterone (Munck et al., 1984), which would be secreted at an increased rate during the initial phase of adaptation to feed deprivation.

Although the effect of feed withdrawal on plasma corticosterone level of a chicken may be influenced by such factors as age, experience, and situation, it is clear

that plasma corticosterone levels normally increase early during feed deprivation. This increase is associated with changes in leucocyte populations and activity, antibody production, and inflammatory responsiveness. Because these changes affect immunological defenses against infection and disease, feed withdrawal evidently changes a hen's ability to resist pathogens. Increase in adrenal secretion of corticosterone has been suggested to confer resistance to cellular pathogens (e.g., bacteria) at the expense of resistance to viral pathogens (Gross, 1983). Along this line, Katanbaf et al. (1987) found increased resistance to *Streptococcus fecalis* in feed-restricted meat-type chickens. However, Holt and Porter (1992) and Holt (1995) found that laying hens' immunity to *Salmonella enteritidis* (nalidixic-acid-resistant strain SE89-8312) was decreased by feed withdrawal. This could be partly due to altered activity of the immune system (Holt, 1992; Kogut et al., 1999) and partly due to the fact that the gut is emptied of its normal contents (Holt et al., 1994). Han and Smyth (1972) reported that resistance of growing chickens to Marek's disease virus (JM strain) was proportionately increased with the level of feed restriction. Feed restriction of meat-type chickens can also increase resistance to *Eimeria tenella*, an intestinal protozoan parasite (Zulkifli et al., 1993).

Feed restriction may increase resistance to pathogenic challenge occurring later in life (Zulkifli et al., 1994 a,b). In these studies, nondwarf chicks that had been restricted from feed during the first week of life were less susceptible than control birds to infection by marble spleen disease virus during heat stress at 5 wk of age. The early-feed-restriction nondwarf chicks also had smaller increases in H:L ratios in response to the heat stress. It is evident that feed deprivation can alter a chicken's preparedness to resist infection. Whether such a change is to the bird's benefit or harm depends on a variety of factors, such as pathogen type, time of pathogen challenge, and genetic stock of the bird.

Plasma Corticosterone and Glucose Metabolism

Although plasma corticosterone and H:L ratios have commonly been measured to interpret stress, it is important to recognize that activation of the hypothalamo-pituitary-adrenal axis in response to feed deprivation has immediate functional significance. Glucocorticoids stimulate hepatic gluconeogenesis, helping to maintain levels of plasma glucose (Munck et al., 1984). The hyperglycemic effect of corticosterone, the glucocorticoid active in domestic fowl, was demonstrated in White Leghorn hens by Greenman and Zarrow (1961). In rats, glucocorticoid influences hepatic phosphorylase activity and glycogen level in the liver during 48 h of starvation, and glucocorticoid is essential for the characteristic compensatory increases of feed consumption and hepatic glycogen content that occur upon refeeding (Bouillon and Berdanier, 1981). Along similar lines, Zulkifli

et al. (1994b) found that young chickens had inferior compensatory growth when refed after metyrapone had been used to block production of corticosterone during feed restriction. In the chicken, the temporary increase of plasma corticosterone during the initial adjustment to feed deprivation probably stimulates glucose output from the liver, helping to ward off hypoglycemia until glucose production from fat catabolism is established. Corticosterone also appears to be important for stimulation of early compensatory growth in the young chicken after at least relatively short periods of feed deprivation.

BEHAVIOR DURING FEED DEPRIVATION

Voluntary anorexia occurs in many species of animals, particularly when feeding would be incompatible with other activities (Mrosovsky and Sherry, 1980). These activities differ among species but include hibernation, egg incubation, territory or harem defense during the breeding season, migration, and molting. Mrosovsky and Sherry (1980) noted that jungle fowl hens incubating eggs restrict feed intake and lose 10 to 20% of their BW during the course of the 20-d incubation period. This reduction of feed intake occurs even when feed is within easy reach of the nesting hen. The pattern of feed intake reduction in relation to BW loss led the authors to suggest that animals exhibiting voluntary anorexia somehow reduce their set-point for defense of BW and adjust their appetite or hunger accordingly. Thus, body mass is allowed to decline due to reduced feeding motivation when the need for feed must be subordinated to another priority.

Savory et al. (1993) observed that broilers on feed restriction would work harder to obtain feed and would eat faster than controls fed ad libitum. Similarly, Petherick and Rutter (1990) found that hens deprived of feed for 43 h were faster than control hens to reach a preset threshold of total force required to open a push-door to gain access to feed. Trained hens that have been deprived of feed will run an alleyway faster to obtain a feed reward (Petherick et al., 1992). Nicol and Guilford (1991) found that hens deprived of feed for a few hours chose to spend increased amounts of time in a novel environment, which the researchers interpreted as being consistent with the notion that feed deprivation elicits increased activity in relation to foraging and exploration. These studies substantiate what is probably common understanding that chickens modify the nature and strength of appetitive feed-related behavior based on how hungry they are.

Duncan and Wood-Gush (1971) tested pairs of hens in several combinations of feed deprivation and food presentation. Aggression by the dominant hen toward the subordinate hen was greatly increased when the hens had been feed deprived for 24 h and then presented with visible feed in a feeder with a clear cover that thwarted their attempts to eat. Feed deprivation followed by no presentation of feed or by the presentation of accessible feed did not lead to increased aggression.

The authors suggested that the aggressive behavior of the thwarted birds indicated frustration. Because aggression was not affected by 24-h food deprivation as long as feed was not visible, the feed-deprived hens might not have been frustrated when they could not see feed. Displacement preening and escape behavior suggestive of frustration or motivational conflict may also be performed by hungry hens when denied access to visible feed (Duncan and Wood-Gush, 1972). Zimmerman et al. (2000a) postulate that the Gakel call of laying hens indicates frustration and found that hens that were feed deprived for 23 h produced Gakel calls in a test situation for which they had been trained to anticipate imminent access to feed.

After hens learned to perform an operant behavior to obtain delivery of feed, they produced Gakel calls when thwarted access to the feed their behavior brought into sight (Zimmerman et al., 2000b). The number of calls increased in accordance with the length of time the hens had been deprived of feed (a maximum of 47 h). The hens' levels of frustration upon being thwarted apparently were modified in accordance with their levels of hunger.

It is not unusual for animals to show a rise in general activity when deprived of feed (Baumeister et al., 1964). Webster (1995) reported that alertness and behavioral activation of hens, as indicated by increased levels of head movement and standing, rose during a 4-d feed withdrawal period. Increased activity has also been observed in feed-restricted broilers (Savory et al., 1992) and mature broiler breeder hens (Hocking et al., 1998). Activity level of feed-restricted growing broiler breeders was found to be associated with suppression of growth rate (Savory et al., 1996; Savory and Lariviere, 2000). This was true regardless of whether the suppression was due to quantitative feed restriction or diet dilution with filler materials. The authors concluded that bulking agents which allow birds to eat increased amounts of food do not reduce hunger and that the general activity level of feed-restricted broiler breeders is closely associated with level of feeding motivation.

Increased non-nutritive pecking, such as pecking structural features of the environment (walls, cage wire, floor) or feathers of other birds, is a typical response of feed-deprived chickens, occurring in situations of feed restriction and total feed withdrawal (Savory and Fisher, 1992; Savory et al., 1992; Webster, 1995, 2000; Hocking et al., 1998). Caged laying hens normally spend 20% or more of their time directing actions into the feeder (Webster and Hurnik, 1990; Webster, 1995). Much of this behavior appears not to involve actual ingestion of feed but resembles foraging-related actions in which the substrate, in this case feed, is flicked to the side or otherwise manipulated with the beak. Therefore, in addition to the likelihood that hunger causes birds to search for food in their environment away from the location of the feeder, some of the increase of non-nutritive pecking may be due to redirection of foraging-related behavior to alternate stimuli in the absence of the

strongly attractive stimulus of a feeder filled with feed. Savory et al. (1996) observed that non-nutritive pecking did not increase when feed restriction was achieved by diet dilution with bulking agents, a situation in which the authors believed that the restricted birds were still hungry.

There has been little systematic study of the behavior of hens during long fasts. Webster (2000) observed a progression of behavior by fasting White Leghorn-type hens that corresponded to the first two phases of fasting outlined by Cherel et al. (1988). The first 3 d appeared to be a time of adjustment. A moderate increase in aggression among fasting birds developed and disappeared during the first day. This sequence may indicate that the hens temporarily experienced frustration as efforts to eat were ineffective but habituated to the situation after a short period. On the second day, fasting hens had elevated behavioral activation and attentiveness as indicated by increased standing, head movement, and non-nutritive pecking. This elevation of activity corresponds to the time at which Etches et al. (1984) found that plasma corticosterone levels peaked during a fast and perhaps involves a stimulatory effect of increased plasma corticosterone on the activity of fasting birds (Astheimer et al., 1992). On the third day, the behavior of the fasting hens appeared to have taken on a pattern consistent with the second phase of fasting as defined by Cherel et al. (1988). This is close to when the rate of BW change of the hens indicated the shift into the second phase of fasting. Standing, head movement, and non-nutritive pecking all declined from d 2, although non-nutritive pecking remained higher than for control hens, and resting behavior increased. Thereafter, the pattern of behavior by fasting hens remained quite stable, being characterized mainly by a gradual rise of resting behavior as the fast progressed. The observations by Simonsen (1979) and Aggrey et al. (1990) of the behavior of hens undergoing feed withdrawal are consistent with the pattern just described. The only aberration to this pattern occurred on d 8 to 10 of the fast when the hens increased preening. Because this happened just before feather pushout took place, the preening probably was due to changes in integumentary stimulation. The hens maintained considerable vigor and agility through the entire period of feed deprivation, which lasted until the hens lost 35% of their prefast BW.

SUMMARY AND CONCLUSIONS

The egg-type laying hen appears to have considerable psycho-physiological ability to cope with long periods without feed. The physiological adjustments to feed deprivation appear to follow the three phases outlined by Cherel et al. (1988) for long-term fasting in birds. During the first phase, which in egg-laying hens may last about 3 d, the hypothalamo-pituitary-adrenal axis is stimulated and plasma corticosterone levels increase. An increased secretion of corticosterone helps the hen to maintain euglycemia during the time needed to engage

metabolic mechanisms to generate adequate supplies of glucose from fat catabolism. The increased plasma corticosterone has additional effects on the production and function of blood leucocytes, and on the performance of other aspects of the hen's immune system. Whether these changes in immune status are beneficial or harmful to the bird appears to depend on the nature and timing of a pathogen challenge. The increased plasma corticosterone also may be associated with increased alertness and food searching behavior. The second phase is quite long in a mature hen. Shimanski (1879) found this phase to extend to 24 d of fasting, but environmental temperature and condition of the hen no doubt affect the actual length of phase 2 in any given case. During phase 2, protein catabolism is minimal and most metabolic energy is derived from fat catabolism. Euglycemia and behavioral vigor are maintained. Plasma corticosterone levels appear to be low during this phase. During phase 3, an increase in protein catabolism occurs, which helps generate glucose to support energy metabolism. The destruction of muscle mass resulting from protein catabolism progressively weakens the bird but vigor is maintained for a number of days. If feed deprivation is allowed to continue, the bird eventually reaches a point when it can no longer be refed and death becomes inevitable.

From the little that is known, the behavior of fasting hens appears to track the physiological phases of fasting, at least for the first two phases. The short period of increased aggression seen by Webster (2000) indicates that the hens in that study may have experienced temporary frustration due to thwarting of feeding behavior early in feed deprivation. Increased alertness and pecking behavior on the second day of fasting could be linked to increased plasma corticosterone levels. If this elevated alertness and pecking is related to hunger, then the decline in these behavioral tendencies, (Webster, 2000), as the hens proceeded to the second phase of fasting may have implications regarding their perception of hunger. The progressive reduction of activity during phase 2 was consistent with minimization of energy utilization during this phase. The hens remained vigorous and agile throughout the portion of phase 2 that was observed. I do not know of any published systematic behavioral studies of the behavior of chickens during phase 3 of fasting, when protein catabolism increases.

It is evident from the array of material covered in this review that the feed withdrawal periods used in properly managed, commercial induced molt programs keep hens within the second phase of physiological adaptation to fasting. It is also evident that a moderate, properly monitored period of feed deprivation need not be a threat to a healthy hen and might even improve aspects of disease resistance and livability. There is, however, the problem that large commercial layer flocks at the end of a production cycle invariably contain a small percentage of birds that are in poor health. These birds apparently succumb at an accelerated rate during

induced molt. From an animal welfare standpoint, it would be best to cull flocks to remove hens in poor condition before feed is withdrawn to induce molt.

A serious effort is underway to find an alternative to the use of feed withdrawal to induce molt in commercial layer flocks. The alternatives that appear most successful thus far in terms of second-cycle performance rely on nutrient restriction by limiting daily feed intake, or by reducing nutrient density or changing nutrient balance of feed offered ad libitum (Rolon et al., 1993; Anderson et al., 2002; Biggs et al., 2002). These alternatives generally cause significant weight loss. If an alternative nutrient restriction program were to delay or prevent a hen from passing into the second physiological stage of fasting, with prolongation of the behavioral states associated with phase 1, i.e., arousal, elevated feed searching behavior and potential for frustration, hen well-being might be made worse than with a single-stage feed withdrawal program. Several studies lend substance to this possibility. For instance, the behavior of qualitatively feed-restricted growing broiler breeder chickens led Savory et al. (1996) and Savory and Lari-viere (2000) to believe that the birds continued to be hungry. Anderson et al. (2002) observed that hens induced to molt by the feeding of a minimum maintenance diet maintained higher levels of aggression during the molt induction period than hens molted by feed withdrawal. It is important to verify that alternative molting procedures actually achieve better hen welfare before they are accepted as replacements for existing commercial induced molting practices.

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